PHYT 6443: Hybrid Therapeutic Interventions
Module 2, Part 3.3: Interventions to Address Primary Mobility Restrictions

Interventions to Address Primary Mobility Deficits due to Joint Restrictions

Objectives

1. List properties of tissues or joints whose primary limitation for mobility is secondary to joint restrictions.
    1. Review principles of inflammation and tissue healing.
    2. Review the stages of tissue healing from Module 1.
    3. Review *tests, measures, and palpatory findings* that will help confirm presence of joint mobility restrictions.
2. Prescribe interventions related to improving mobility in tissues and joints that are restricted by joint mobility.
    1. Define the components of intervention prescriptions (mode, intensity, duration, and frequency) for joint mobilizations based on the purpose.
    2. List characteristics of the different phases of tissue healing for non-operative and post-operative management.
    3. List goals, indications, weaknesses, precautions, and contraindications for joint mobilizations.
    4. Provide at least 2 examples of how to progress and regress selected interventions.
        1. List signs/symptoms that would indicate need to progress or regress intervention.
3. Assess efficacy of joint mobilizations with objective tests and measures learned in PT Exam Skills, as well as palpatory findings. (Test, perform intervention with appropriate prescription, re-test)
4. Review examples of joint mobilizations to target mobility restrictions in joints, as well as neuromuscular effects.

What is "Joint Mobility Assessment," and What are Joint Mobilizations?

Joint Mobility Assessment

When we assess joint mobility, we are assessing the **joint play**, which is "involuntary movement that is present in all synovial joints. Voluntary movement is dependent on the integrity of joint play." (Menell)

Therefore, appropriate amounts of joint play support normal gross movement for joints. Joint play should only be about 1/8 of an inch, so it is very small overall. With appropriate practice and training, you can still feel about 1/8 of an inch of movement. Fair warning, just like surface anatomy and palpation, it is a skill/art that must be practiced. The lab videos will show you how to assess, and then turn those assessments into treatments. We will continue to practice and reinforce these skills throughout the remainder of the program.

When a joint is in dysfunction (doesn't move how it normally should), it can lead to either hypermobility or hypomobility. What is important to remember is that if joint play is lost, it is a mechanical problem that requires a mechanical solution. The change in accessory motion, by the very definition of accessory motion, cannot be restored with muscle activation.

Joint Mobilizations

Mobilizations are passive, skilled manual therapy techniques applied to joints using physiological or accessory motions.

- Mobilizations are graded according to the amplitude and depth of mobilizations, often using the Grade I-IV approach.
- Grades I/II are better for pain modulation and fluid imbibition. Grades III/IV are better for improving joint mobility.

Grading Joint Mobilizations

So what do I mean by Grade I/II versus Grade III/IV. These are called Maitland Mobilization grades. These grades refer to the amount of mobility in any given direction for a joint. You have to know what Grade IV is for an individual's joint in a specific direction to know how to implement Grade I, II, or III. This does not mean get a running start. Rather, gradually work into Grade IV with your assessment. To help you with this, I am posting a video description of the mobilizations and a picture to help with visualization. I will repeat these descriptions later in the module to try to help reinforce and tie the information together.

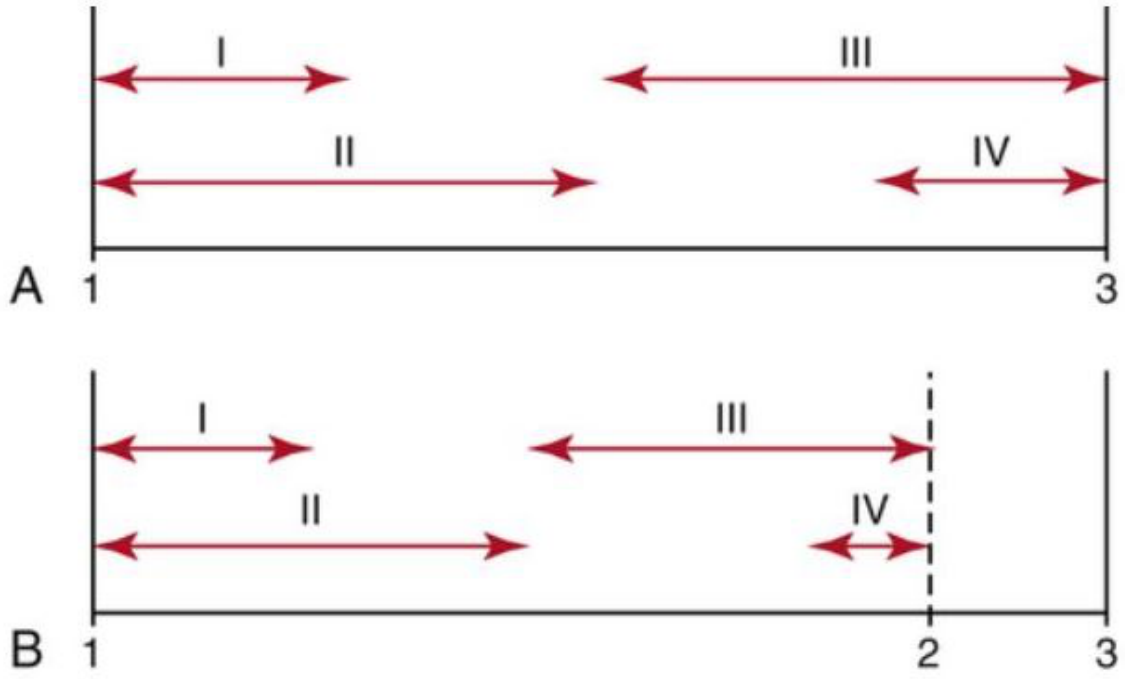


FIG. 15-4 **A,** Grades of oscillations used in manual therapy. **B,** Grades of oscillations used in manual therapy in relation to a joint with limited motion. *1,* Starting position of movement; *2,* point of limitation of movement; *3,* anatomical limit of movement. (From Gould JA: *Orthopaedic and sports physical therapy*, ed 2, St Louis, 1990, Mosby.)

https://clinicalgate.com/concepts-of-joint-mobilization/

Joint Manipulations

Manipulation (High-velocity thrust): high velocity, short amplitude technique performed at the end of the pathological limit of the joint.

- Intended to alter positional relationships, snap adhesions, or stimulate joint receptors.
- Best used when there is a loss of joint mobility in order to restore mobility at that segment.
- May hear a little snap, crackle, pop, but this is just nitrogen gas rapidly being released from a joint.

We will learn specific manipulations for joints beginning next year. I want you to get good at mobilizations before we turn to manipulations.

These may also be called Grade V mobilizations in the clinic when you are on rotation.

Did you know? Not all states permit Physical Therapists to perform joint manipulations. It is a big point of contention in our practice act, and it is an important reason we remain active with our professional organizations. It helps protect our ability to perform techniques like these and other techniques.

But how does it work?

What is most important for you to know is the effects of joint mobilizations. There are 3 main effects you need to understand. The effects are listed below. Click over the effects to learn more about how the joint mobilizations address these topics.

Effects of Joint Mobilizations

Performing Joint Mobility Assessment

How to Assess Joint Mobility

The following is a list of Mennell's Rules for Joint Play Testing, which is directly quoted from the textbook *Orthopedic Physical Assessment* by Magee DJ.

- "The patient should be relaxed and fully supported.
- The examiner should be relaxed and should use a firm but comfortable grasp.
- One joint should be examined at a time.
- One movement should be examined at a time.
- The unaffected side should be tested first.
- One articular surface is stabilized, while the other surface is moved.
- Movements must be normal and not forced.
- Movements should not cause undue discomfort." (Magee 2017)

In summary, assessment of joint

mobility will require more work on you as a therapist. The patient should not be helping the assessment. If the patient "helps," they risk activating muscles, which would impair your joint mobility assessment. Also, the patient should be supported, with the joint in as loose-packed of a position as possible. This will help you determine if it is truly the joint that is restricted vs. another tissue. So, what do we need to do? Love 'em, touch 'em, squeeze 'em! Look at the image to the right. Consider the rules above, and identify which rules the therapist is following. Then, scroll over the caption under the image to check your answers.

Scroll over the image above to see how joint play testing rules are being used in this image.

Arthrokinomatics

Arthrokinematics is the primary principle that fuels joint mobilizations. This means that you need to be aware of all the arthrokinematics within the body to effectively apply joint mobilizations. For example, if you push something in the wrong direction, you could make their condition worse. Typically, if you know the "glide" component of the arthrokinematics, that is the direction that you will push.

Example: The glenohumeral joint rolls superior and glides inferior with shoulder ABduction with normal motion. Thus, if there is a joint issue and the person is found have limitations with Abduction, you as a PT would look to push the join in the direction that it would normally glide AKA in an inferior direction.

The good news is that you do not need to memorize every single joint because all joints follow the typical arthrokinematics principles of roll-glide-spin based on their structure. Knowing that, you only need to know the joint structure and then apply your typical convex-concave/other rules.

Physiopedia has a nice page (https://www.physio-pedia.com/Arthrokinematics) that they have devoted to arthrokinomatics. While this topic is mostly covered in your movement science 1 class, we rely heavily on this information. Your written exam will test you on arthrokinematics, joint alignment, concave/convex rules, and open packed and closed packed positions. In the below table, blocked position = closed packed; relaxed = open packed

* *JOINT POSITIONS* *

joint	blocked position	relaxed	capsular pattern
hip	max extension, inter. rotation, abduction	flexion-30°, abd-30° small external rotation	flexion, internal rotation, abduction
knee	max extension, external rotation	flexion-25°	flexion, extension
tibiofibular	max knee extension, dorsal foot flexion	knee semiflexion, semiplantar foot flexion	pain when joint stressed
talocrural	max dorsiflexion	plantar flexion-10°	Plantar flexion; dorsiflexion
subtalar	supination	middle position	limitation of varus range of movement
midtarsal	dorsal flexion	semiplantar flexion	dorsiflexion, plantar flexion, adduction, medial rotation
first MTP	max extension	semiflexion	extension, flexion
second to fifth MTP	max extension	semiflexion	variable
IP	max extension	semiflexion	flexion, extension
glenohumeral	max abduction and external rotation	abd-55°, add-30°, internal rotation	lateral rotation, abduction, medial rotation
sternoclavicular	depresion of shoulder	semielevation of shoulder	pain at extreme range of movement
acromioclavicular	shoulder elevation	shoulder depresion	pain at extreme range of movement
humeroulnar	max extension and supination	flexion-70°, supinatio-10°	flexion, extension
radiohumeral	flexion-90°, supination-5°	max. extension, supination	flexion, extension, supination, pronation
proximal radioulnar	supination-5°	supination-35°, flexion of elbow-70°	supination, pronation
distal radioulnar	supination-5°	supinatio-10°	pain at extremes of rotation
wrist	max extension and radial deviation	semi dorsal flexion	flexion & extension equally limited
trapeziometacarpal	max opposition	semi: flex, ext, abd, add	abduction, extension
MCP and IP	max flexion	semiflexion	flexion, extension
temporomandibular	closed	semiopening	opening
occipitoatlanto	max extension	semiflexion	extension & side flexion equally limited
cervical spine	max extension	semiflexion	side flexion & rotations equally limited, extension
thoracic spine	max extension	semiflexion	side flexion & rotation equally limited, extension
lumbar spine	max extension	semiflexion	side flexion & rotation equally limited, extension
SI, symphysis pubis, and sacrococcygeal	pelvic inclination	pelvic reclination	pain when joints stressed

table taken from physiopedia.

Application of the Concave-Convex Rule

When assessing joint mobility and performing joint mobilizations, always think about the convex/concave rules. You are inducing the glide component of the technique. On joints that are concave on convex, whatever direction you are moving a bone is in the same direction of osteokinematic movement. On joints that are convex on concave, whatever direction you are moving a bone is in the opposite direction of osteokinematic movement.

For example, let's consider the glenohumeral joint. The humeral head is convex, and the glenoid fossa is concave. For shoulder abduction, the humeral head rolls superior and glides inferior. When you assess inferior glide of the humeral head, it is associated with glenohumeral abduction.

Documenting Joint Mobility Findings

If you assess joint mobility and findings, you would document these findings in the objective portion of the test. You would typically document the joint and the direction of mobility assessed. For example, you could document "Posterior glide of the tibia on the femur, grade ___." The following is a 7-point system developed by Stanley Paris. It is the most commonly recognized form of documenting the degree of joint mobility.

Grade	Interpretation	Meaning
0	Ankylosed	No detectable movement in the joint. Do not try to mobilize or manipulate these joints.
1	Considerable restriction	Significant decrease in expected range. Significant resistance to movement. Grade III/IV mobilizations will be beneficial.
2	Slight restriction	Notable decrease in expected range. Grade III/IV mobilizations will be beneficial.
3	Normal	Normal mobility detected for that joint compared to opposite side, age, gender, occupation, recreational activities.
4	Slight increase	Some increase in expected range. Less than normal resistance to movement. Focus on stabilization/motor control exercises, but you may be able to perform Grade I/II mobilizations for pain relief or fluid imbibition.
5	Considerable increase	Excessive range, but eventually restricted by capsular and ligamentous structures. Focus on stabilization/motor control exercises, but you may be able to perform Grade I/II mobilizations for pain relief or fluid imbibition.
6	Unstable	Excessive range, without the capsular and ligamentous restraint. As a clinical pearl, if you say something is unstable, it should require surgical or some other medical intervention to increase stability.

Turning Examination Findings into Treatment

What did you find?

What directions did the joint not move as well? How does that relate to a patient's loss of mobility? For example, if a patient has a loss of hip flexion, did they have a loss of hip posterior and inferior glides? Did they have decreased distraction? The assessment findings turn into your treatment mobilizations.

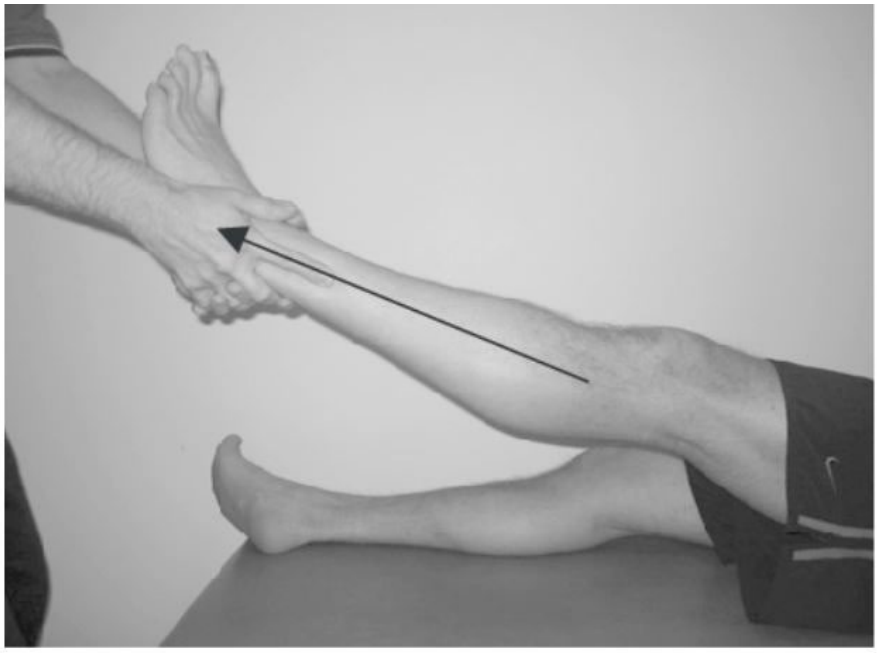
In tissues that are acutely irritated, have high levels of pain, or early on in the rehabilitation processes, you will want to plan to start joint mobilizations in the open-packed joint position. As the patient's pain decreases, joint mobility increases, or time has passed, you can progress joint mobilizations into more of a closed-packed position to take advantage of the tissue wind-up.

How to Mobilize

Once you decide 1) the direction that you want the bone segment to move in, 2) whether the patient would best benefit from the open-packed or non-open-packed position, and 3) the intensity of the joint mobilization, you are ready to mobilize.

First, stabilize the patient so that they are comfortable and that you are only getting movement at the desired joint. Sometimes, that involves you manually stabilizing the rest of the patient's body, and sometimes, gravity, mobilization belts, and other people are used. The key is to isolate the intended joint so that you are sure your force is in a very targeted direction.

Line your body up to meet the needs of the joint mobilization. Sometimes, this is perpendicular to the joint, like Pieter did in picture on page 4 of this module for the lumbar spine. Sometimes, this is parallel to the joint, like you see in the hip distraction mobilization to the right.



Long-axis nonthrust mobilization/manipulation of the hip in 15° to 30° abduction and 15° to 30° flexion.

Speaking of distraction...

In addition to our standard glide mobilizations (anterior, posterior, medial, lateral, superior, and/or inferior), we can also have spin mobilizations. These combine

several directions of movement.

Joints also have distraction, which is the separation of 2 joint surfaces directly away from each other. These distraction mobilizations are beneficial when a joint has had too much compression, or when the patient's pain levels are too high to tolerate other mobilizations.

Joints also have compression, which is bringing the 2 joint surfaces closer to each other. These will be practiced in your various course when cases are presented.

Review of Joint Mobilization Grades

Watch the video below for a review of joint mobilization grades. Practice these joint mobilization grades on your own MCP joint.

Mode, Intensity, Duration, Frequency for Joint Mobilizations

Acute injury or high levels of pain

- Mode: Joint mobilizations, usually in open-packed position
- Intensity: Grade I/II for fluid imbibition. May be able to progress to Grade III/IV within treatment session, but must proceed with caution.
- Duration: Total should be about 100 oscillations at about 100 beats per minute (bpm). This can be broken down into 6-10 sets of 6-10 repetitions.
- Frequency: Daily as needed to decrease pain or imbibe good fluid.

Subacute/Chronic injuries with less pain

- Mode: Joint mobilizations, progressing from open-packed position towards closed-packed positions
- Intensity: Grade III/IV to improve joint mobility
- Duration: Total should be about 100 oscillations at about 100 beats per minute (bpm). This can be broken down into 6-10 sets of 6-10 repetitions.
- Frequency: Daily as needed to improve joint mobility.

Safety Things

Contraindications to Joint Mobilizations

- Presence of serious pathology that the mobilization may adversely affect
- Inability of the therapist to perform the technique
- Fracture in the region
- Ligament rupture in that region
- No working hypothesis
- Multi-level nerve root pathology (indicates something more serious)
- Worsening neurological function (the patient needs to see another provider)
- Unremitting, severe, non-mechanical pain
- Unremitting night pain that prevents the patient from sleeping
- Empty end feel and severe multidirectional spasm
- Presence of upper motor neuron lesions that is not being addressed additionally by another provider. Special note: If a patient has been assessed for the upper motor neuron lesion and it is deemed safe to mobilize, then proceed.

Precautions to Joint Mobilizations

- Any pathology that can weaken bone (i.e. osteoporosis, cancer)
- Certain medications, such as steroidal drugs or anticoagulants
- Pregnancy (risk of miscarriage, specifically at 19 weeks)
- Inflammatory diseases
- Depleted general health
- Patient's inability to relax
- Physique (do NOT put yourself at risk trying to achieve a certain mobilization. If you get hurt. then you cannot help other patients who would benefit from your skill set.)
- Undiagnosed pain
- Lack of patient consent
- If the joint is not "giving"
- Adverse reaction to previous manual therapy
- Acute disc herniation/prolapse
- Pain with psychological overlay
- When the specific joint mobilizations or palpation reproduces distal symptoms
- Chronic pain and fibromyalgia since

This image is here to grab your attention. You need to know these precautions and contraindications. You can expect to see exam questions on these, but straight-up rote memorization and in case study form.

the patients would benefit more from a pain science approach • Emotionally dependent patients Used with permission from Kroon P, Krutchowsky T. Introduction to Lumbar Spine I. The Manual Therapy Institute, PLLC; 2022.	

Muscle Energy Techniques (METs): Where joint mobilizations and muscle activation combines!

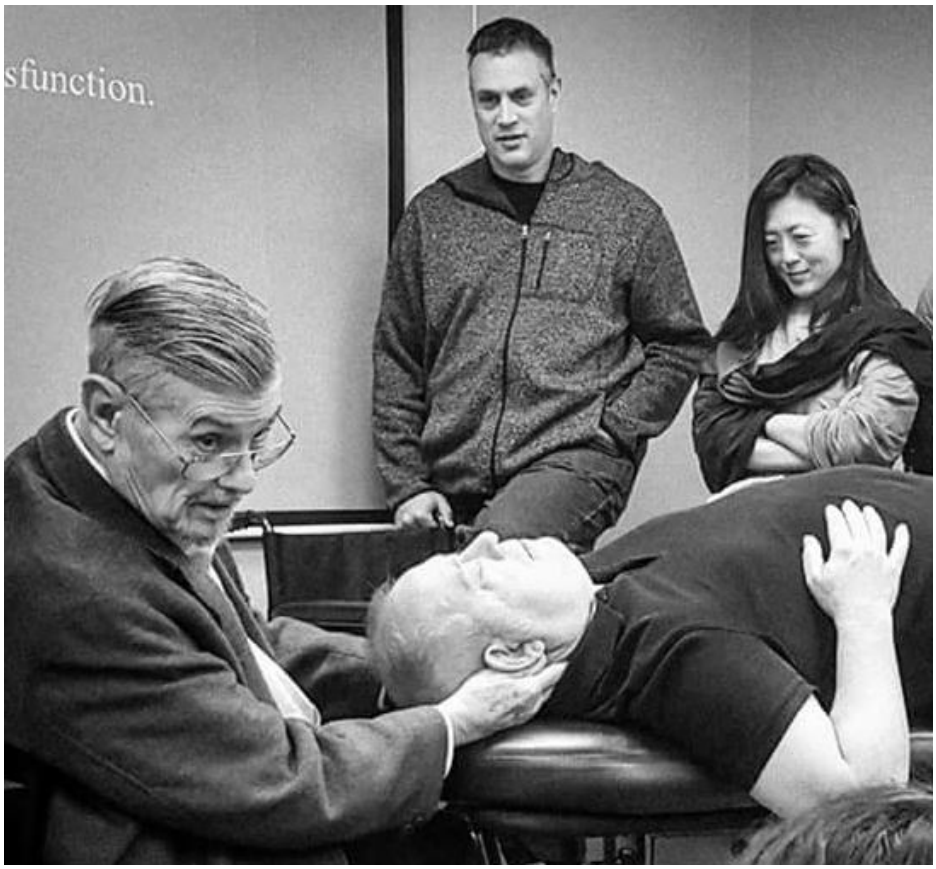
What are musculoskeletal METs?

A muscle energy technique (MET) refers to any mobilization that involves the voluntary use of the patient's muscles. Think about it like PNF combined with joint mobilizations (see videos of soft tissue lower extremity techniques).

How METs came to be?

The technique was developed by Dr. Fred Mitchell, Sr., DO (osteopath) and Dr. Fred Mitchell, Jr., DO around 1950. The thought process behind this procedure was that muscles become hypertonic, which can restrict the osteokinematic function of the motion segment. To restore joint mobility, the technique involves accurate localization of the motion barrier and specific muscle activation to encourage a joint to move either into the barrier of motion or away from the barrier of motion. Osteopaths in the United States currently practice and treat more like MDs in the United States in order to keep up their practice and medical privileges. Osteopaths in Europe and a few decades ago in the USA practice more like PTs do now.

The image to the right was taken in 2016 in Ontario, Canada. These techniques are still new when we think about the science and art of Physical Therapy. The techniques will take some practice because it involves gentle patient handling, but they are very useful tools to aid your clinical planning.



Example of an MET Asking the Agonist to Work

Directions: For side glide right. Can be done in neutral, semi-flexion/ extension. Side glide head to right up to restriction barrier, which creates left side bending. Instruct patient to push their head gently to their right side. The therapist manually resists for 6-8 seconds. Then pick up slack in right side gliding. Perform 3-4 repetitions. Notice how the right cervical spine muscles are being asked to contract in a lengthened position with this technique.

How do you know which muscle to use?

Asking the Antagonist to Work

There are some METs that you will perform where you want to contract the muscles that are antagonists to the muscle with dysfunction. The idea is that contraction of the antagonist muscle will cause the muscle in dysfunction to relax. With this kind of technique, it is very important that you palpate the muscle in dysfunction to ensure that the force is not so great that you cause a stabilization reaction. As we go through a few key muscle energy techniques in this course and in Ortho, I will help you figure out whether you are asking the agonist or antagonist to work.

Asking the Agonist to Work

Likewise, you can perform an MET where you ask the patient to contract the muscle in dysfunction from a lengthened position. The idea behind this is the force produced by the muscle will activate the Golgi tendon organs, which, as you know from Neuroscience, causes reflexive inhibition to the muscle.

We will practice identifying the agonist and antagonist more in immersion.

Review of Dosing

"A muscle energy technique is applied from a precisely controlled position, in a specific direction and against a distinctly executed counterforce. The counterforce is in the region of ounces, not pounds. For a muscle energy technique to be successful, proper localization is key. This makes MET's not the easiest techniques to do as palpation skills need to be fairly well advanced."

- Kroon P, Kruchowsky T. Introduction to Cervical Spine I. The Manual Therapy Institute, PLLC; 2022.

Clinical Summary

METs are excellent tools to have in your back pocket. Practice them, and use them frequently, especially if the end feel of a joint is predominately muscular or the patient has difficulty relaxing.

Regular practice of joint mobilizations and METs is critical for their success. The wind-up has to be very localized with precise resistance provided to the patient to ensure the correct muscle is activating without other muscles "jumping in to help." It takes confidence and skills with palpation. You may not feel anything in the beginning but as your hands develope, you will be able to sense small changes in tissue movement.